
BPS-17 Lecturer Mathematics AJKPSC/FPSC/PPSC/KPSC

Study Plan & MCQ Bank

Competitive Examination Preparation

PSC — BPS-17 Mathematics Lecturer

Topics Covered

[regular] Differential Geometry	[regular] Vector Analysis
[regular] Tensor Analysis	[regular] Number Theory
[regular] Real Analysis	[regular] Complex Analysis
[regular] Linear Algebra	[regular] Differential Equations
[regular] Numerical Methods	[regular] Calculus

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1 Study Plan Overview

PSC Exam Tips

AJKPSC BPS-17 Exam Pattern (Mathematics Lecturer)

- **Paper:** Written + MCQs (Combined objective/subjective)
- **Total Marks:** 100 (MCQ portion typically 50–60 marks)
- **Time:** 2–3 hours
- **Syllabus:** BSc/MSc level pure and applied mathematics
- **Repeated Topics:** Vector Analysis, Real Analysis, Differential Equations, Number Theory, Linear Algebra, Complex Analysis

Important Formulae — Recommended Study Schedule (8 Weeks)

Week	Topic	Priority
1	Number Theory + Linear Algebra	High
2	Vector Analysis	High
3	Differential Geometry	High
4	Tensor Analysis	Medium
5	Real Analysis	High
6	Complex Analysis	High
7	Differential Equations	High
8	Numerical Methods + Revision	Medium

2 Differential Geometry

Key Concepts — Differential Geometry

1. **Curves in $\mathbb{R}^3$:** A space curve is given by $\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle$.

2. **Unit Tangent Vector:**

$$\mathbf{T} = \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|}$$

3. **Curvature:**

$$\kappa = \frac{|\mathbf{T}'|}{|\mathbf{r}'|} = \frac{|\mathbf{r}' \times \mathbf{r}''|}{|\mathbf{r}'|^3}$$

4. **Torsion:**

$$\tau = \frac{(\mathbf{r}' \times \mathbf{r}'') \cdot \mathbf{r}'''}{|\mathbf{r}' \times \mathbf{r}''|^2}$$

5. **Frenet–Serret Formulae:**

$$\mathbf{T}' = \kappa \mathbf{N}, \quad \mathbf{N}' = -\kappa \mathbf{T} + \tau \mathbf{B}, \quad \mathbf{B}' = -\tau \mathbf{N}$$

where $\mathbf{N}$ = principal normal, $\mathbf{B} = \mathbf{T} \times \mathbf{N}$ = binormal.

6. **Arc Length:**

$$s = \int_a^b |\mathbf{r}'(t)| dt$$

7. **First Fundamental Form** of a surface $\mathbf{r}(u, v)$:

$$ds^2 = E du^2 + 2F du dv + G dv^2$$

where $E = \mathbf{r}_u \cdot \mathbf{r}_u$, $F = \mathbf{r}_u \cdot \mathbf{r}_v$, $G = \mathbf{r}_v \cdot \mathbf{r}_v$.

8. **Gaussian Curvature:**

$$K = \frac{LN - M^2}{EG - F^2}$$

9. **Mean Curvature:**

$$H = \frac{EN - 2FM + GL}{2(EG - F^2)}$$

2.1 MCQs — Differential Geometry

Q1. The curvature of a straight line is:

- (A) ∞
- (B) 1
- (C) 0
- (D) -1

Answer & Explanation**Correct Answer: (C)**

A straight line has no bending at any point, so its curvature $\kappa = 0$. Recall $\kappa = |\mathbf{r}' \times \mathbf{r}''|/|\mathbf{r}'|^3$; for a line $\mathbf{r}'' = \mathbf{0}$, giving $\kappa = 0$.

Q2. For a circle of radius r , the curvature is:

- (A) r
- (B) r^2
- (C) $1/r$
- (D) $1/r^2$

Answer & Explanation**Correct Answer: (C)**

For a circle of radius r , curvature $\kappa = 1/r$. A tighter circle (smaller r) has larger curvature — this matches our intuition.

Q3. The torsion of a plane curve is:

- (A) 1
- (B) κ
- (C) ∞
- (D) 0

Answer & Explanation**Correct Answer: (D)**

A plane curve lies in a fixed plane, so its binormal $\mathbf{B}$ is constant. Since $\mathbf{B}' = -\tau\mathbf{N}$ and $\mathbf{B}' = \mathbf{0}$, we get $\tau = 0$.

Q4. The Frenet–Serret formula $\mathbf{N}' = -\kappa\mathbf{T} + \tau\mathbf{B}$ relates:

- (A) Curvature and torsion of a curve
- (B) Surface area to curvature
- (C) Geodesics on a surface
- (D) Gaussian and mean curvature

Answer & Explanation**Correct Answer: (A)**

The Frenet–Serret formulae form a system describing how the TNB frame rotates along the curve, relating $\mathbf{T}$, $\mathbf{N}$, $\mathbf{B}$ through curvature κ and torsion τ .

Q5. If the Gaussian curvature $K > 0$, the surface is called:

- (A) Hyperbolic
- (B) Parabolic
- (C) Elliptic
- (D) Flat

Answer & Explanation

Correct Answer: (C)

Surfaces with $K > 0$ are elliptic (e.g., sphere). Surfaces with $K < 0$ are hyperbolic (e.g., saddle). Surfaces with $K = 0$ are flat/parabolic.

Q6. The arc length of the helix $\mathbf{r}(t) = (a \cos t, a \sin t, bt)$ from $t = 0$ to $t = 2\pi$ is:

- (A) $2\pi a$
- (B) $2\pi b$
- (C) $2\pi\sqrt{a^2 + b^2}$
- (D) $2\pi(a + b)$

Answer & Explanation

Correct Answer: (C)

$|\mathbf{r}'(t)| = |(-a \sin t, a \cos t, b)| = \sqrt{a^2 + b^2}$. Arc length $= \int_0^{2\pi} \sqrt{a^2 + b^2} dt = 2\pi\sqrt{a^2 + b^2}$.

Q7. The binormal vector $\mathbf{B}$ is defined as:

- (A) $\mathbf{T} + \mathbf{N}$
- (B) $\mathbf{T} \times \mathbf{N}$
- (C) $\mathbf{N} \times \mathbf{T}$
- (D) $|\mathbf{T}'|$

Answer & Explanation

Correct Answer: (B)

The binormal vector $\mathbf{B} = \mathbf{T} \times \mathbf{N}$. It is perpendicular to both $\mathbf{T}$ and $\mathbf{N}$, completing the orthonormal TNB (Frenet) frame.

Q8. A surface has $E = 1$, $F = 0$, $G = 1$. The metric coefficients correspond to:

- (A) A sphere
- (B) A cylinder
- (C) A flat plane
- (D) A paraboloid

Answer & Explanation

Correct Answer: (C)

When $E = G = 1$ and $F = 0$, the first fundamental form is $ds^2 = du^2 + dv^2$, which is the standard Euclidean metric — corresponding to a flat plane.

PSC Exam Tips

PSC Exam Tips — Differential Geometry

- Memorise the Frenet-Serret formulae — they appear almost every year.
- Know the formula for curvature of a circle ($\kappa = 1/r$) and straight line ($\kappa = 0$).
- Torsion $\tau = 0$ for any plane curve — a frequently tested fact.
- Practice: curvature and torsion of helices are very common MCQ candidates.
- Remember: Gaussian curvature $K = LN - M^2/(EG - F^2)$ — know this formula by heart.

3 Vector Analysis

Key Concepts — Vector Analysis

1. Del Operator:

$$\nabla = \frac{\partial}{\partial x} \hat{\mathbf{i}} + \frac{\partial}{\partial y} \hat{\mathbf{j}} + \frac{\partial}{\partial z} \hat{\mathbf{k}}$$

2. Gradient, Divergence, Curl:

$$\text{grad } f = \nabla f, \quad \text{div } \mathbf{F} = \nabla \cdot \mathbf{F}, \quad \text{curl } \mathbf{F} = \nabla \times \mathbf{F}$$

3. Laplacian:

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$$

4. Important Identities:

$$\begin{aligned} \nabla \times (\nabla f) &= \mathbf{0} \quad (\text{curl of gradient} = 0) \\ \nabla \cdot (\nabla \times \mathbf{F}) &= 0 \quad (\text{div of curl} = 0) \\ \nabla \cdot (f\mathbf{F}) &= f(\nabla \cdot \mathbf{F}) + \mathbf{F} \cdot \nabla f \\ \nabla \times (f\mathbf{F}) &= f(\nabla \times \mathbf{F}) + (\nabla f) \times \mathbf{F} \end{aligned}$$

5. Gauss's Divergence Theorem:

$$\oint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_V (\nabla \cdot \mathbf{F}) dV$$

6. Stokes' Theorem:

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$$

7. Green's Theorem:

$$\oint_C (P dx + Q dy) = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

8. Irrotational and Solenoidal Fields:

- $\mathbf{F}$ is irrotational if $\nabla \times \mathbf{F} = \mathbf{0}$
- $\mathbf{F}$ is solenoidal if $\nabla \cdot \mathbf{F} = 0$

3.1 MCQs — Vector Analysis

Q1. If $f = x^2 + y^2 + z^2$, then ∇f at $(1, 1, 1)$ is:

- (A) (1, 1, 1)
- (B) (2, 2, 2)
- (C) (0, 0, 0)
- (D) (3, 3, 3)

Answer & Explanation**Correct Answer: (B)** $\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right) = (2x, 2y, 2z)$. At (1, 1, 1): $\nabla f = (2, 2, 2)$.**Q2.** For any scalar function f , $\nabla \times (\nabla f)$ equals:

- (A) $\nabla^2 f$
- (B) $\mathbf{0}$
- (C) ∇f
- (D) f

Answer & Explanation**Correct Answer: (B)**

The curl of the gradient of any scalar field is always the zero vector: $\nabla \times (\nabla f) = \mathbf{0}$. This is a fundamental vector identity.

Q3. A vector field $\mathbf{F}$ is solenoidal if:

- (A) $\nabla \times \mathbf{F} = 0$
- (B) $\nabla \cdot \mathbf{F} = 0$
- (C) $\nabla f = \mathbf{F}$
- (D) $\nabla^2 \mathbf{F} = 0$

Answer & Explanation**Correct Answer: (B)**

Solenoidal (divergence-free) means $\nabla \cdot \mathbf{F} = 0$. Irrotational means $\nabla \times \mathbf{F} = \mathbf{0}$. Do not confuse these two.

Q4. Gauss's Divergence Theorem converts:

- (A) A line integral to a surface integral
- (B) A surface integral to a volume integral
- (C) A volume integral to a line integral
- (D) A line integral to an area integral

Answer & Explanation**Correct Answer: (B)**

Gauss's theorem: $\oint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_V (\nabla \cdot \mathbf{F}) dV$ — it converts a closed surface integral (2D boundary) to a volume integral (3D interior).

Q5. If $\mathbf{F} = (y^2z, x^2z, xyz)$, then $\nabla \cdot \mathbf{F}$ is:

- (A) 0
- (B) xyz
- (C) xy
- (D) $x^2 + y^2 + z^2$

Answer & Explanation**Correct Answer: (C)**

$$\nabla \cdot \mathbf{F} = \frac{\partial}{\partial x}(y^2z) + \frac{\partial}{\partial y}(x^2z) + \frac{\partial}{\partial z}(xyz) = 0 + 0 + xy = xy.$$

Q6. Stokes' theorem relates:

- (A) Volume integral to surface integral
- (B) Surface integral to line integral
- (C) Line integral to double integral
- (D) Double integral to triple integral

Answer & Explanation**Correct Answer: (B)**

Stokes' theorem: $\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$. It converts a line integral around a closed curve to a surface integral of curl.

Q7. The directional derivative of f in the direction of unit vector $\hat{\mathbf{u}}$ is:

- (A) $|\nabla f|$
- (B) $\nabla f \cdot \hat{\mathbf{u}}$
- (C) $\nabla f \times \hat{\mathbf{u}}$
- (D) $\nabla^2 f$

Answer & Explanation**Correct Answer: (B)**

The directional derivative $D_{\hat{\mathbf{u}}}f = \nabla f \cdot \hat{\mathbf{u}}$. The gradient ∇f gives the direction of maximum increase; projecting onto $\hat{\mathbf{u}}$ gives the rate in that direction.

Q8. For $\mathbf{F} = x^2\hat{i} + y^2\hat{j} + z^2\hat{k}$, is $\mathbf{F}$ irrotational?

- (A) Yes, always
- (B) No, never
- (C) Only at origin
- (D) Only if $x = y = z$

Answer & Explanation

Correct Answer: (A)

$\nabla \times \mathbf{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial_x & \partial_y & \partial_z \\ x^2 & y^2 & z^2 \end{vmatrix} = \mathbf{0}$ since each cross-derivative is zero. So $\mathbf{F}$ is irrotational everywhere.

PSC Exam Tips

PSC Exam Tips — Vector Analysis

- The two golden identities: $\nabla \times (\nabla f) = \mathbf{0}$ and $\nabla \cdot (\nabla \times \mathbf{F}) = 0$ — always MCQ targets.
- Know when each integral theorem (Green's, Stokes', Gauss's) applies.
- Solenoidal $\Leftrightarrow$ divergence $= 0$. Irrotational $\Leftrightarrow$ curl $= 0$. Get these right.
- Practice computing ∇f , $\nabla \cdot \mathbf{F}$, and $\nabla \times \mathbf{F}$ quickly.

4 Tensor Analysis

Key Concepts — Tensor Analysis

1. Einstein Summation Convention: A repeated index (one up, one down) implies summation:

$$a_i b^i = \sum_{i=1}^n a_i b^i$$

2. Kronecker Delta:

$$\delta_j^i = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

3. Levi-Civita Symbol:

$$\varepsilon_{ijk} = \begin{cases} +1 & \text{if } (i, j, k) \text{ is an even permutation of } (1, 2, 3) \\ -1 & \text{if } (i, j, k) \text{ is an odd permutation} \\ 0 & \text{if any index is repeated} \end{cases}$$

4. Contravariant and Covariant Tensors:

- Contravariant (upper index): transforms as $\bar{A}^i = \frac{\partial \bar{x}^i}{\partial x^j} A^j$
- Covariant (lower index): transforms as $\bar{A}_i = \frac{\partial x^j}{\partial \bar{x}^i} A_j$

5. Metric Tensor:

$$ds^2 = g_{ij} dx^i dx^j$$

The metric tensor g_{ij} raises and lowers indices:

$$A^i = g^{ij} A_j, \quad A_i = g_{ij} A^j$$

6. Christoffel Symbols:

$$\Gamma_{ij}^k = \frac{1}{2} g^{kl} \left(\frac{\partial g_{li}}{\partial x^j} + \frac{\partial g_{lj}}{\partial x^i} - \frac{\partial g_{ij}}{\partial x^l} \right)$$

7. Covariant Derivative:

$$A^i_{;j} = \frac{\partial A^i}{\partial x^j} + \Gamma_{jk}^i A^k$$

8. Riemann Curvature Tensor:

$$R^l_{ijk} = \frac{\partial \Gamma^l_{ik}}{\partial x^j} - \frac{\partial \Gamma^l_{ij}}{\partial x^k} + \Gamma^l_{jm} \Gamma^m_{ik} - \Gamma^l_{km} \Gamma^m_{ij}$$

4.1 MCQs — Tensor Analysis

Q1. A tensor of rank 0 is a:

- (A) Vector
- (B) Scalar
- (C) Matrix
- (D) Dyadic

Answer & Explanation

Correct Answer: (B)

A tensor of rank (order) 0 has no free indices — it is a scalar. A tensor of rank 1 is a vector, and rank 2 is a matrix.

Q2. The Kronecker delta δ_j^i is a:

- (A) Contravariant tensor
- (B) Covariant tensor
- (C) Mixed tensor of type (1,1)
- (D) Rank-3 tensor

Answer & Explanation

Correct Answer: (C)

δ_j^i has one upper index and one lower index, making it a mixed tensor of type (1, 1) (once contravariant, once covariant).

Q3. Under the Einstein summation convention, δ_i^i equals:

- (A) 0
- (B) 1
- (C) n (dimension of space)
- (D) ∞

Answer & Explanation

Correct Answer: (C)

$\delta_i^i = \sum_{i=1}^n \delta_i^i = \sum_{i=1}^n 1 = n$. In 3D space, $\delta_i^i = 3$.

Q4. The Levi-Civita symbol ε_{123} equals:

- (A) -1
- (B) 0
- (C) $+1$
- (D) 3

Answer & Explanation

Correct Answer: (C)

$(1, 2, 3)$ is an even (identity) permutation of $(1, 2, 3)$, so $\varepsilon_{123} = +1$.

Q5. A flat (Euclidean) space has Riemann curvature tensor:

- (A) $R^l_{ijk} = 1$
- (B) $R^l_{ijk} = g_{ij}$
- (C) $R^l_{ijk} = 0$
- (D) $R^l_{ijk} = \Gamma^l_{ij}$

Answer & Explanation

Correct Answer: (C)

The Riemann tensor measures intrinsic curvature. For flat Euclidean space, all Christoffel symbols vanish (in Cartesian coordinates), so $R^l_{ijk} = 0$.

Q6. The number of independent components of a symmetric tensor A_{ij} in n dimensions is:

- (A) n^2
- (B) $n(n-1)/2$
- (C) $n(n+1)/2$
- (D) n

Answer & Explanation

Correct Answer: (C)

A symmetric tensor $A_{ij} = A_{ji}$ has n diagonal + $\binom{n}{2} = n(n-1)/2$ off-diagonal pairs = $n(n+1)/2$ total.

Q7. The covariant derivative of the metric tensor $g_{ij;k}$ is:

- (A) Γ_{ijk}
- (B) g_{ij}
- (C) 0
- (D) R_{ijk}

Answer & Explanation**Correct Answer: (C)**

By the fundamental theorem of Riemannian geometry (metric compatibility), $g_{ij;k} = 0$. The covariant derivative of the metric tensor always vanishes.

Q8. Contraction of a tensor A^{ij} over its indices gives:

- (A) A tensor of rank 2
- (B) A tensor of rank 0 (scalar)
- (C) A tensor of rank 3
- (D) A vector

Answer & Explanation**Correct Answer: (B)**

Contracting both indices of a rank-2 contravariant tensor A^{ij} (setting $i = j$ and summing) reduces the rank by 2, yielding a scalar A^{ii} (the trace).

PSC Exam Tips**PSC Exam Tips — Tensor Analysis**

- Einstein summation: a repeated index (up + down) = sum. Never write three of the same index.
- Remember: $\delta_i^i = n$, $\varepsilon_{ijk}\varepsilon^{ijk} = 6$ in 3D.
- Metric compatibility: $g_{ij;k} = 0$ — very commonly tested.
- Riemann tensor = 0 $\Leftrightarrow$ flat space — know this equivalence.
- Tensor rank terminology: rank 0 = scalar, rank 1 = vector, rank 2 = matrix.

5 Number Theory

Key Concepts — Number Theory

1. **Divisibility:** $a \mid b$ means $b = ka$ for some integer k .

2. **GCD and LCM:**

$$\gcd(a, b) \cdot \text{lcm}(a, b) = |ab|$$

3. **Euclidean Algorithm:** $\gcd(a, b) = \gcd(b, a \bmod b)$, continue until remainder is 0.

4. **Bezout's Identity:** $\exists x, y \in \mathbb{Z}$ such that $\gcd(a, b) = ax + by$.

5. **Prime Numbers:** A prime $p > 1$ has exactly two divisors: 1 and p .

6. **Fundamental Theorem of Arithmetic:** Every integer $n > 1$ can be expressed uniquely as a product of primes.

7. **Euler's Totient Function $\phi(n)$:** $\phi(n)$ = number of integers in $\{1, \dots, n\}$ coprime to n .

$$\phi(p) = p - 1, \quad \phi(p^k) = p^{k-1}(p - 1), \quad \phi(mn) = \phi(m)\phi(n) \text{ if } \gcd(m, n) = 1$$

8. **Fermat's Little Theorem:** If p is prime and $\gcd(a, p) = 1$:

$$a^{p-1} \equiv 1 \pmod{p}$$

9. **Euler's Theorem:** If $\gcd(a, n) = 1$:

$$a^{\phi(n)} \equiv 1 \pmod{n}$$

10. **Chinese Remainder Theorem (CRT):** If $m_1, m_2, \dots, m_k$ are pairwise coprime, the system $x \equiv a_i \pmod{m_i}$ has a unique solution modulo $M = m_1 m_2 \cdots m_k$.

11. **Wilson's Theorem:** p is prime $\Leftrightarrow (p-1)! \equiv -1 \pmod{p}$.

12. **Congruence Properties:** If $a \equiv b \pmod{m}$, then $a + c \equiv b + c$ and $ac \equiv bc \pmod{m}$.

5.1 MCQs — Number Theory

Q1. What is $\gcd(48, 18)$?

- (A) 3
- (B) 6
- (C) 9
- (D) 12

Answer & Explanation

Correct Answer: (B)

Using the Euclidean algorithm: $\gcd(48, 18) = \gcd(18, 12) = \gcd(12, 6) =$

$$\gcd(6, 0) = 6.$$

Q2. By Fermat's Little Theorem, $2^{10} \pmod{11}$ equals:

- (A) 0
- (B) 1
- (C) 2
- (D) 10

Answer & Explanation

Correct Answer: (B)

11 is prime and $\gcd(2, 11) = 1$. By Fermat: $2^{10} \equiv 2^{11-1} \equiv 1 \pmod{11}$.

Q3. $\phi(12)$ equals:

- (A) 3
- (B) 4
- (C) 6
- (D) 8

Answer & Explanation

Correct Answer: (B)

$12 = 2^2 \cdot 3$. $\phi(12) = \phi(4) \cdot \phi(3) = 2 \cdot 2 = 4$. The integers coprime to 12 in $\{1, \dots, 12\}$ are: 1, 5, 7, 11 — exactly 4.

Q4. Wilson's Theorem states that $(p-1)! \equiv -1 \pmod{p}$ if and only if p is:

- (A) Even
- (B) Composite
- (C) Prime
- (D) A perfect square

Answer & Explanation

Correct Answer: (C)

Wilson's Theorem is an *iff* characterisation of primality: $(p-1)! \equiv -1 \pmod{p} \Leftrightarrow p$ is prime.

Q5. The number of primes less than or equal to 20 is:

- (A) 6
- (B) 7
- (C) 8
- (D) 9

Answer & Explanation**Correct Answer: (C)**Primes ≤ 20 : 2, 3, 5, 7, 11, 13, 17, 19 — exactly 8.**Q6.** If $\gcd(a, b) = 1$, then a and b are called:

- (A) Twin primes
- (B) Coprime (relatively prime)
- (C) Perfect numbers
- (D) Mersenne primes

Answer & Explanation**Correct Answer: (B)**Two integers with $\gcd = 1$ are called *coprime* or *relatively prime*. They share no common prime factor.**Q7.** The solution to $x \equiv 2 \pmod{3}$ and $x \equiv 3 \pmod{5}$ by CRT is:

- (A) $x \equiv 5 \pmod{15}$
- (B) $x \equiv 8 \pmod{15}$
- (C) $x \equiv 11 \pmod{15}$
- (D) $x \equiv 13 \pmod{15}$

Answer & Explanation**Correct Answer: (B)**We need $x = 3k + 2$ and $3k + 2 \equiv 3 \pmod{5} \Rightarrow 3k \equiv 1 \pmod{5} \Rightarrow k \equiv 2 \pmod{5}$.
So $k = 5m + 2$, $x = 15m + 8$. Thus $x \equiv 8 \pmod{15}$.**Q8.** Which of the following is a Mersenne prime?

- (A) $2^4 - 1 = 15$
- (B) $2^5 - 1 = 31$
- (C) $2^6 - 1 = 63$
- (D) $2^8 - 1 = 255$

Answer & Explanation**Correct Answer: (B)**

A Mersenne prime is a prime of the form $2^p - 1$. $31 = 2^5 - 1$ is prime. $15 = 3 \times 5$, $63 = 9 \times 7$, $255 = 5 \times 51$ are composite.

PSC Exam Tips**PSC Exam Tips — Number Theory**

- Euclidean algorithm is the fastest way to find GCD — practice it until it is automatic.
- Fermat's Little Theorem: $a^{p-1} \equiv 1 \pmod{p}$ — used to compute large powers mod p quickly.
- Euler's totient $\phi(p) = p - 1$ for prime p ; compute $\phi(n)$ using prime factorisation.
- Wilson's Theorem is the only simple primality test expressible as a congruence.
- CRT: remember it only applies when all moduli are pairwise coprime.

6 Real Analysis

Key Concepts — Real Analysis

1. **Sequences:** (a_n) converges to L if $\forall \varepsilon > 0, \exists N$ such that $n > N \Rightarrow |a_n - L| < \varepsilon$.

2. **Series convergence tests:**

- **Ratio test:** $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = L$. Converges if $L < 1$, diverges if $L > 1$.
- **Root test:** $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = L$. Same criterion.
- **Comparison test:** $0 \leq a_n \leq b_n$ and $\sum b_n$ converges $\Rightarrow \sum a_n$ converges.
- **Integral test:** $\sum a_n$ and $\int_1^\infty f(x) dx$ behave alike (for positive decreasing f).
- **Alternating Series (Leibniz):** $\sum (-1)^n a_n$ converges if $a_n \searrow 0$.

3. **Continuity:** f is continuous at c iff $\lim_{x \rightarrow c} f(x) = f(c)$.

4. **Uniform Continuity:** $\forall \varepsilon > 0, \exists \delta > 0$ (independent of x) such that $|x - y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon$.

5. **Differentiability:** $f'(c) = \lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h}$.

6. **Mean Value Theorem:** If f is continuous on $[a, b]$ and differentiable on (a, b) :

$$\exists c \in (a, b) \text{ such that } f'(c) = \frac{f(b) - f(a)}{b - a}$$

7. **Riemann Integrability:** f is Riemann integrable on $[a, b]$ iff it is bounded and continuous a.e.

8. **Cauchy Sequence:** (a_n) is Cauchy if $\forall \varepsilon > 0, \exists N: m, n > N \Rightarrow |a_m - a_n| < \varepsilon$.
In $\mathbb{R}$: Cauchy $\Leftrightarrow$ convergent.

6.1 MCQs — Real Analysis

Q1. Which sequence converges?

- (A) $a_n = n$
- (B) $a_n = (-1)^n$
- (C) $a_n = 1/n$
- (D) $a_n = \sin n$

Answer & Explanation

Correct Answer: (C)

$a_n = 1/n \rightarrow 0$ as $n \rightarrow \infty$. The others: $n \rightarrow \infty$ (diverges), $(-1)^n$ oscillates, $\sin n$ does not converge.

Q2. The series $\sum_{n=1}^{\infty} \frac{1}{n}$ is:

- (A) Convergent
- (B) Divergent
- (C) Absolutely convergent
- (D) Conditionally convergent

Answer & Explanation

Correct Answer: (B)

The harmonic series $\sum 1/n$ diverges (shown by the integral test: $\int_1^{\infty} dx/x = \infty$). This is a fundamental result in analysis.

Q3. By the Mean Value Theorem, for $f(x) = x^2$ on $[1, 3]$, the value of c is:

- (A) 1
- (B) 2
- (C) 3
- (D) $\sqrt{3}$

Answer & Explanation

Correct Answer: (B)

$f'(c) = \frac{f(3)-f(1)}{3-1} = \frac{9-1}{2} = 4$. Since $f'(x) = 2x$, we get $2c = 4 \Rightarrow c = 2$.

Q4. Every Cauchy sequence in $\mathbb{R}$ is:

- (A) Monotone
- (B) Bounded but not convergent
- (C) Convergent
- (D) Divergent

Answer & Explanation

Correct Answer: (C)

$\mathbb{R}$ is a complete metric space: Cauchy $\Leftrightarrow$ convergent in $\mathbb{R}$. This is the Completeness Axiom.

Q5. A function f is uniformly continuous on $(0, 1)$ if f is:

- (A) Unbounded
- (B) Continuous on $[0, 1]$
- (C) Differentiable
- (D) Monotone

Answer & Explanation

Correct Answer: (B)

A function continuous on a closed bounded interval $[a, b]$ is uniformly continuous there (Heine-Cantor theorem). This does not hold for open intervals in general.

PSC Exam Tips

PSC Exam Tips — Real Analysis

- Harmonic series $\sum 1/n$ diverges; $\sum 1/n^2$ converges — know both.
- MVT formula: $f'(c) = [f(b) - f(a)]/(b - a)$ — find c by substitution.
- Cauchy = convergent in $\mathbb{R}$ (completeness). This is tested conceptually.
- Ratio test: $L < 1$ converges, $L > 1$ diverges, $L = 1$ inconclusive.

7 Complex Analysis

Key Concepts — Complex Analysis

- Complex Number:** $z = x + iy$, $|z| = \sqrt{x^2 + y^2}$, $\bar{z} = x - iy$.
- Cauchy–Riemann Equations:** For $f = u + iv$ to be analytic:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

- Cauchy's Integral Formula:**

$$f^{(n)}(a) = \frac{n!}{2\pi i} \oint_C \frac{f(z)}{(z-a)^{n+1}} dz$$

- Residue Theorem:**

$$\oint_C f(z) dz = 2\pi i \sum \text{Res}(f, a_k)$$

- Residue at a simple pole:**

$$\text{Res}(f, a) = \lim_{z \rightarrow a} (z-a)f(z)$$

- Laurent Series:** $f(z) = \sum_{n=-\infty}^{\infty} c_n(z-a)^n$

- Singularities:**

- Removable: $\lim_{z \rightarrow a} f(z)$ exists
- Pole of order m : $c_{-m} \neq 0$, $c_n = 0$ for $n < -m$
- Essential: infinitely many negative-power terms

- Liouville's Theorem:** Every bounded entire function is constant.

- Morera's Theorem:** If $\oint_C f(z) dz = 0$ for every closed contour, then f is analytic.

7.1 MCQs — Complex Analysis

Q1. For $f(z) = z^2$, verify: f is:

- (A) Nowhere analytic
- (B) Analytic only at origin
- (C) Analytic everywhere (entire)
- (D) Analytic only for $\text{Im}(z) = 0$

Answer & Explanation**Correct Answer: (C)**

$z^2 = (x + iy)^2 = x^2 - y^2 + 2xyi$. Check C-R: $u_x = 2x = v_y$, $u_y = -2y = -v_x$. Both hold everywhere. So f is entire.

Q2. The function $f(z) = \bar{z}$ is:

- (A) Entire
- (B) Analytic at $z = 0$ only
- (C) Nowhere analytic
- (D) Analytic for $\text{Re}(z) > 0$

Answer & Explanation**Correct Answer: (C)**

For $f = \bar{z}$: $u = x$, $v = -y$. C-R requires $u_x = v_y$: $1 = -1$ — false everywhere. So $\bar{z}$ is nowhere analytic.

Q3. The residue of $f(z) = \frac{1}{z(z-1)}$ at $z = 1$ is:

- (A) -1
- (B) 0
- (C) 1
- (D) 2

Answer & Explanation**Correct Answer: (C)**

$z = 1$ is a simple pole. $\text{Res}(f, 1) = \lim_{z \rightarrow 1} (z - 1) \cdot \frac{1}{z(z-1)} = \lim_{z \rightarrow 1} \frac{1}{z} = 1$.

Q4. Liouville's Theorem states that a bounded entire function must be:

- (A) Zero
- (B) Periodic
- (C) Constant
- (D) Analytic

Answer & Explanation**Correct Answer: (C)**

Liouville's Theorem: if f is entire (analytic on all of $\mathbb{C}$) and bounded ($|f(z)| \leq M$),

then f is a constant.

Q5. The Cauchy-Riemann equations are *necessary* for f to be:

- (A) Continuous
- (B) Differentiable (analytic) at a point
- (C) Bounded
- (D) Real-valued

Answer & Explanation

Correct Answer: (B)

The C-R equations are *necessary* for complex differentiability at a point. Combined with the existence of continuous partial derivatives, they are also sufficient.

PSC Exam Tips

PSC Exam Tips — Complex Analysis

- C-R equations: $u_x = v_y$ and $u_y = -v_x$ — memorise them exactly (sign on the second!).
- $\bar{z}$ is nowhere analytic — a very common MCQ trap.
- Residue at a simple pole: $\text{Res}(f, a) = \lim_{z \rightarrow a} (z - a)f(z)$.
- Liouville's Theorem is tested conceptually almost every year.
- Entire = analytic everywhere on $\mathbb{C}$ (e.g., polynomials, e^z , $\sin z$, $\cos z$).

8 Linear Algebra

Important Formulae — Key Linear Algebra Results

$$\det(AB) = \det(A)\det(B), \quad \det(A^{-1}) = \frac{1}{\det(A)}, \quad \det(A^T) = \det(A)$$

$$\text{rank}(A) + \text{nullity}(A) = n \quad (\text{Rank-Nullity Theorem})$$

Eigenvalue equation: $A\mathbf{v} = \lambda\mathbf{v}$, found from $\det(A - \lambda I) = 0$.

Cayley-Hamilton: Every matrix satisfies its own characteristic equation.

Gram-Schmidt: Orthogonalises a set of linearly independent vectors.

8.1 MCQs — Linear Algebra

Q1. The rank of $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{pmatrix}$ is:

- (A) 0
- (B) 1
- (C) 2
- (D) 3

Answer & Explanation

Correct Answer: (B)

All rows are multiples of $(1, 2, 3)$: row 2 = 2(row 1), row 3 = 3(row 1). So there is exactly one independent row: $\text{rank}(A) = 1$.

Q2. If A is a 3×3 matrix with $\det(A) = 5$, then $\det(2A)$ is:

- (A) 10
- (B) 20
- (C) 40
- (D) 80

Answer & Explanation

Correct Answer: (C)

$\det(kA) = k^n \det(A)$ for an $n \times n$ matrix. So $\det(2A) = 2^3 \cdot 5 = 40$.

Q3. The eigenvalues of $A = \begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix}$ are:

- (A) 1 and 3
- (B) 3 and 3
- (C) 0 and 3
- (D) 1 and 1

Answer & Explanation

Correct Answer: (B)

$\det(A - \lambda I) = (3 - \lambda)^2 = 0 \Rightarrow \lambda = 3$ (repeated). This is a defective (non-diagonalisable) matrix.

Q4. The Rank-Nullity theorem states: for $A : \mathbb{R}^n \rightarrow \mathbb{R}^m$:

- (A) $\text{rank}(A) = \text{nullity}(A)$
- (B) $\text{rank}(A) + \text{nullity}(A) = m$
- (C) $\text{rank}(A) + \text{nullity}(A) = n$
- (D) $\text{rank}(A) \cdot \text{nullity}(A) = n$

Answer & Explanation

Correct Answer: (C)

Rank-Nullity: $\text{rank}(A) + \text{nullity}(A) = n$, where n is the number of *columns* (domain dimension). Not m (the codomain dimension).

Q5. A matrix is singular if and only if:

- (A) Its trace is zero
- (B) One eigenvalue is zero
- (C) It is symmetric
- (D) It is upper triangular

Answer & Explanation

Correct Answer: (B)

A matrix is singular $\Leftrightarrow \det(A) = 0 \Leftrightarrow$ at least one eigenvalue is zero (since $\det(A) = \prod \lambda_i$).

PSC Exam Tips

PSC Exam Tips — Linear Algebra

- $\det(kA) = k^n \det(A)$ for $n \times n$ — the n exponent is commonly forgotten.
- Eigenvalues of a triangular matrix are its diagonal entries.

- Rank-Nullity: the n on the right is the number of columns of A .
- Singular $\Leftrightarrow \det = 0 \Leftrightarrow$ not invertible $\Leftrightarrow$ zero eigenvalue.

9 Differential Equations

Important Formulae — ODE Methods and Results

First Order ODE types:

- **Separable:** $\frac{dy}{dx} = f(x)g(y) \Rightarrow \int \frac{dy}{g(y)} = \int f(x) dx$
- **Linear:** $\frac{dy}{dx} + P(x)y = Q(x)$; integrating factor $\mu = e^{\int P dx}$
- **Exact:** $M dx + N dy = 0$ with $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$
- **Bernoulli:** $y' + P(x)y = Q(x)y^n$; substitution $v = y^{1-n}$

Second Order Linear ODE $y'' + py' + qy = f(x)$:

- Characteristic equation: $m^2 + pm + q = 0$
- Two real roots $m_1 \neq m_2$: $y_h = c_1 e^{m_1 x} + c_2 e^{m_2 x}$
- Repeated root m : $y_h = (c_1 + c_2 x) e^{mx}$
- Complex roots $\alpha \pm \beta i$: $y_h = e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x)$

Laplace Transform:

$$\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t) dt$$

Key pairs: $\mathcal{L}\{1\} = \frac{1}{s}$, $\mathcal{L}\{e^{at}\} = \frac{1}{s-a}$, $\mathcal{L}\{\sin \omega t\} = \frac{\omega}{s^2 + \omega^2}$, $\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}$

9.1 MCQs — Differential Equations

Q1. The ODE $\frac{dy}{dx} = \frac{x}{y}$ is:

- (A) Linear
- (B) Exact
- (C) Separable
- (D) Bernoulli

Answer & Explanation

Correct Answer: (C)

Rearrange: $y dy = x dx$ — variables separate. Integrating: $\frac{y^2}{2} = \frac{x^2}{2} + C$, i.e., $y^2 - x^2 = C$.

Q2. The integrating factor for $y' + \frac{2}{x}y = x^2$ is:

- (A) e^{2x}
- (B) x^2
- (C) e^x
- (D) $1/x^2$

Answer & Explanation

Correct Answer: (B)

$P(x) = 2/x$. Integrating factor $\mu = e^{\int (2/x) dx} = e^{2 \ln x} = x^2$.

Q3. The general solution to $y'' - 5y' + 6y = 0$ is:

- (A) $c_1 e^{2x} + c_2 e^{3x}$
- (B) $c_1 e^{-2x} + c_2 e^{-3x}$
- (C) $(c_1 + c_2 x) e^{3x}$
- (D) $e^x (c_1 \cos 2x + c_2 \sin 2x)$

Answer & Explanation

Correct Answer: (A)

Characteristic equation: $m^2 - 5m + 6 = 0 \Rightarrow (m - 2)(m - 3) = 0 \Rightarrow m = 2, 3$.

General solution: $y = c_1 e^{2x} + c_2 e^{3x}$.

Q4. $\mathcal{L}\{e^{3t}\}$ equals:

- (A) $\frac{1}{s+3}$
- (B) $\frac{1}{s-3}$
- (C) $\frac{1}{s^2+9}$
- (D) $\frac{1}{s^2-9}$

Answer & Explanation

Correct Answer: (B)

Using the standard formula: $\mathcal{L}\{e^{at}\} = \frac{1}{s-a}$. With $a = 3$: $\mathcal{L}\{e^{3t}\} = \frac{1}{s-3}$ (valid for $s > 3$).

Q5. The condition for $M dx + N dy = 0$ to be an exact equation is:

- (A) $M = N$
- (B) $\frac{\partial M}{\partial x} = \frac{\partial N}{\partial y}$
- (C) $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$
- (D) $MN = 1$

Answer & Explanation

Correct Answer: (C)

Exactness requires $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ — both are equal to $\frac{\partial^2 F}{\partial x \partial y}$ for some potential function F .

PSC Exam Tips

PSC Exam Tips — Differential Equations

- Memorise all four 1st-order types and their solution methods.
- Characteristic equation roots $\rightarrow$ solution form: two real, repeated, complex. Know all three.
- Laplace table: $\mathcal{L}\{1\} = 1/s$, $\mathcal{L}\{e^{at}\} = 1/(s - a)$, $\mathcal{L}\{t^n\} = n!/s^{n+1}$.
- Exactness test: $M_y = N_x$ — *cross* derivatives.

10 Numerical Methods

Important Formulae — Key Numerical Formulae

Bisection Method: $x_{n+1} = \frac{a_n + b_n}{2}$ (halves interval each step).

Newton-Raphson Method:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Converges quadratically near a simple root.

Trapezoidal Rule:

$$\int_a^b f(x) dx \approx \frac{h}{2} [f(x_0) + 2f(x_1) + \cdots + 2f(x_{n-1}) + f(x_n)], \quad h = \frac{b-a}{n}$$

Simpson's $\frac{1}{3}$ Rule (n must be even):

$$\int_a^b f(x) dx \approx \frac{h}{3} [f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + \cdots + f(x_n)]$$

Lagrange Interpolation:

$$P(x) = \sum_{i=0}^n f(x_i) \prod_{j \neq i} \frac{x - x_j}{x_i - x_j}$$

Euler's Method: $y_{n+1} = y_n + h f(x_n, y_n)$

10.1 MCQs — Numerical Methods

Q1. Newton-Raphson method has ____ order of convergence:

- (A) Linear
- (B) Quadratic
- (C) Cubic
- (D) Logarithmic

Answer & Explanation

Correct Answer: (B)

The Newton-Raphson method converges quadratically (order 2) near a simple root — the error roughly squares at each step. This is its main advantage over bisection (order 1).

Q2. Simpson's $1/3$ rule requires the number of subintervals n to be:

- (A) Odd
- (B) Even
- (C) A multiple of 3
- (D) Any positive integer

Answer & Explanation

Correct Answer: (B)

Simpson's $1/3$ rule is derived from pairs of intervals, so it requires n to be even. Simpson's $3/8$ rule requires n to be a multiple of 3.

Q3. The bisection method always converges if:

- (A) $f(a) > 0$ and $f(b) > 0$
- (B) $f(a)$ and $f(b)$ have opposite signs
- (C) f is differentiable on $[a, b]$
- (D) $f(a) = 0$

Answer & Explanation

Correct Answer: (B)

By the Intermediate Value Theorem, if $f(a)$ and $f(b)$ have opposite signs and f is continuous, a root exists in (a, b) . The bisection method is guaranteed to converge in this case.

PSC Exam Tips

PSC Exam Tips — Numerical Methods

- Newton-Raphson: quadratic convergence, requires $f'(x_n) \neq 0$.
- Simpson's rule: n must be even, weights are $1, 4, 2, 4, 2, \dots, 4, 1$.
- Trapezoidal weights: $1, 2, 2, \dots, 2, 1$.
- Bisection: slow but always converges; Newton-Raphson: fast but may diverge.

11 Quick Reference Card

Must-Know Formulae at a Glance

Differential Geometry

- $\kappa = |\mathbf{r}' \times \mathbf{r}''|/|\mathbf{r}'|^3$
- $\mathbf{T}' = \kappa\mathbf{N}$, $\mathbf{B}' = -\tau\mathbf{N}$
- Plane curve: $\tau = 0$
- Circle: $\kappa = 1/r$

Vector Analysis

- $\nabla \times (\nabla f) = \mathbf{0}$
- $\nabla \cdot (\nabla \times \mathbf{F}) = 0$
- Solenoidal: $\nabla \cdot \mathbf{F} = 0$
- Irrotational: $\nabla \times \mathbf{F} = \mathbf{0}$

Number Theory

- Fermat: $a^{p-1} \equiv 1 \pmod{p}$
- Wilson: $(p-1)! \equiv -1 \pmod{p}$
- $\phi(p) = p-1$; $\phi(p^k) = p^{k-1}(p-1)$
- $\gcd(a, b) \cdot \text{lcm}(a, b) = |ab|$

Tensor Analysis

- $\delta_i^i = n$ (dimension)
- $g_{ij;k} = 0$ (metric compatibility)
- Flat: $R_{ijk}^l = 0$
- Rank 0=scalar, 1=vector, 2=matrix

Complex Analysis

- C-R: $u_x = v_y$, $u_y = -v_x$
- $\bar{z}$: nowhere analytic
- Liouville: bounded entire $\Rightarrow$ constant
- Residue (simple pole): $\lim_{z \rightarrow a} (z - a)f(z)$

Differential Equations

- I.F. = $e^{\int P dx}$
- Exact: $M_y = N_x$
- Char. eq. for 2nd order ODE
- $\mathcal{L}\{e^{at}\} = 1/(s - a)$

*

Best of Luck for AJKPSC BPS-17 Examination!

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